OBSERVATION

ANNOUNCEMENT

Assignment: Role Play of a Scientific Figure

Deliverable 2: Script – due Feb 6th, 11:59 pm

Please incorporate some of the concepts we have discussed in our philosophy of science module.

I will be looking for at least ***1 to 2 examples*** connecting what we have learned to the work and legacy of the scientific figure you have chosen.

The Plan!

- Recap
- Underdetermination cont.
- Observation

RECAP

Underdetermination

- The logical structure of an explanation and testing is ***never*** of the *form that the evidence entails a theory*: Rather, it is the other way around.
  - A *theory entails its evidence*, and because of this, *observation of the evidence cannot stand as clear confirmation of the theory*.
  - Nor can the evidence provide unambiguous falsification since, as we have seen (e.g., solar neutrino problem, OPERA case), *realistic cases of testing are of the form, theory* ***plus auxiliaries*** *entail the evidence*.

Underdetermination

To put it bluntly,

- the observations of the world, that is, the reports of *manifest* (observable) phenomena, are *not sufficiently informative* to single out the one true theory of what is going on behind the *appearances*.
  - **Note**: Explanation and testing are strict and crucial discriminators of scientific theories in that a theory is unacceptable if it is in *no way explanatory* and *does not cooperate with other theories to make true predictions*.
  - The comparison of a theory with observations weeds out lots of potential theories …

Underdetermination

- The problem of **underdetermination** is that this process doesn't weed out enough, since many competing theories can account *equally well* for the evidence.
- ***The evidence alone cannot trim the pool of applicants down to the single theory that is the true theory.***

Underdetermination

Last time!

- As long as an alternative account is conceptually possible (and it always is), the observational data alone can give the theory we have in mind at best a fifty-fifty chance of being true.
- In fact, though, the **problem of underdetermination** is much worse than this because,
    - for any finite amount of data (things explained, tests passed), there will always be *any number* of theories that are empirically equivalent over that data.

Underdetermination

Theory Underdetermination:

Which theory we choose is *underdetermined* by the evidence. (That is, other theories will also fit the evidence.)

Underdetermination

- The pool of empirically adequate theories can be trimmed toward the truth only in cooperation with ***internal standards (virtues***).
  - Internal virtues:

    1. Entrenchment
    2. Explanatory cooperation
    3. Testability
    4. Generality
    5. Simplicity

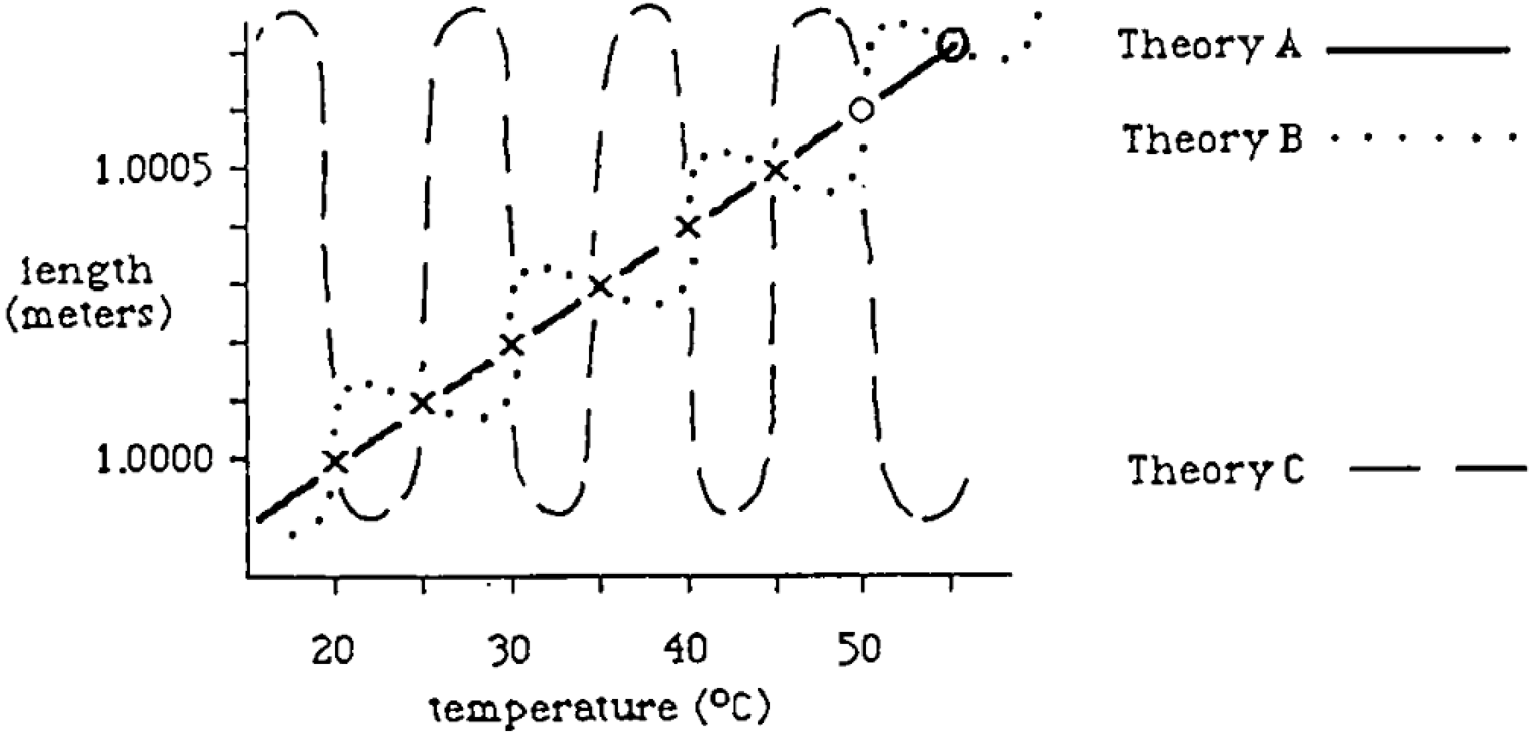
Underdetermination

The heated-bar example

Theory A explains all the observed events and correctly predicts the future test at 50 degrees.

But so does theory B.

Theory A is preferable on the standards of *simplicity.*



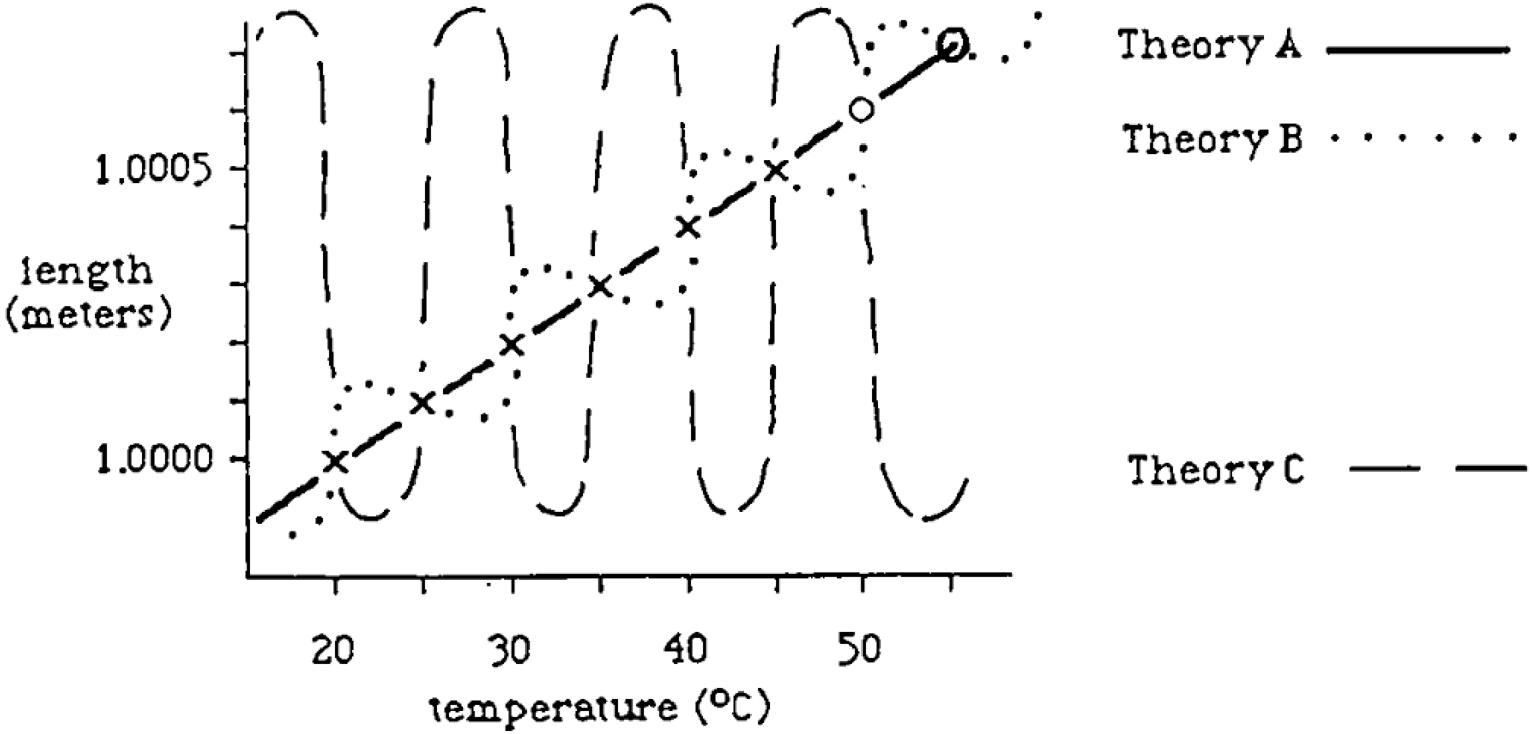
Underdetermination

Last tim

The example demonstrates that the world ***seems*** as if theory A is true.

But based on the observation, it appears as well as if B is true, and clearly both cannot be right.

This forces a close look at the question of the inference from evidence to theory, the inference from "seems as if" to "is."



Underdetermination

1. Instrumentalism:

- This brand of antirealism claims that scientific theories are not intended to be evaluated as being true or false.
  - Theories are useful or not, applicable or not, but they are not *objectively* true or false.
- They are only ***instruments*** we use to guide our thoughts and beliefs about the ***world we can observe***.
  - Theories are intellectual tools for purpose of prediction of future phenomena and organization of observations.

Underdetermination

1. Instrumentalism:

- evaluates scientific theories in terms of ***usefulness*** and ***applicability***, rather than in terms of truth.

- *Empirical adequacy* – efficient organization and successful prediction of phenomena – is the goal of theorizing.
  - Equally empirical adequate theories are equally acceptable. As there is no one true hammer, there is no one true theory of gravity.

Underdetermination

1. Instrumentalism:

- On this view, theories function as pedagogical tools for experts.
  - For example, science teachers might use a conceptual aid to help their students understand the physical world. The atoms in a molecule, for instance, can be visualized as being connected by inner springs.

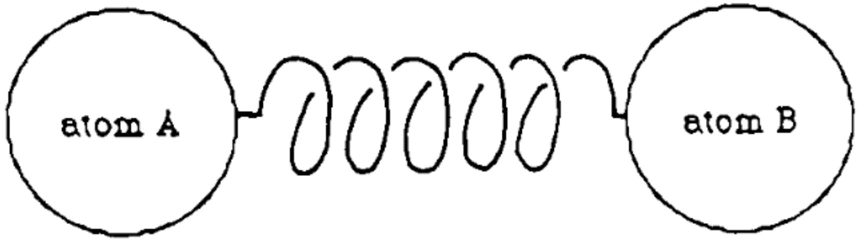


Figure 5.2. A model of the diatomic molecule *AB*, with the binding force represented as a spring.

Last time!

1. Instr

- On t

This conceptualization of springs between atoms helps us to get the calculations right …

But no science teacher means to say that there really ***are*** tiny springs between the atoms.

The claim about the real world is only that it is ***as if*** there are such springs.

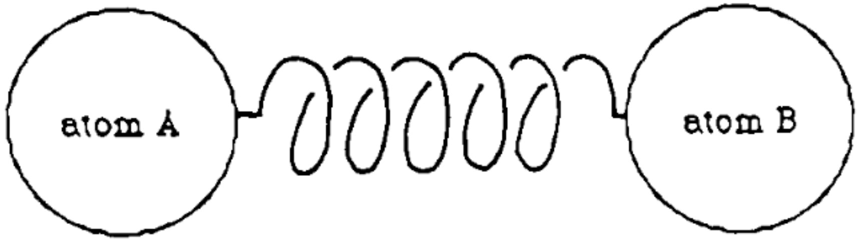


Figure 5.2. A model of the diatomic molecule AB, with the binding force represented as a spring.

Underdetermination

1. Instrumentalism:

- Similarly, unobservable entities (such as electrons, quarks, atoms) ***should not*** be taken ***literally***, and so it would be inappropriate to call them true or to call them false.
- Thus, we should stop endorsing talking about unobservables like electrons and atoms as true.

Underdetermination

- One of the reasons why one might subscribe to the view of **instrumentalism** is to distance oneself from the mystical and the occult (e.g., unobservable forces as found in astrology).
  - There are no such things as electrons, quarks, or atoms, just as there is no such thing as God.
- This happens for the price of really understanding what is happening behind the scenes in the realm of unobservables – **instrumentalism** ignores the scientific surprise and curiosity about the unobservable.
  - However, something *causes* the world as we see it, meaning, observable objects are constituted of something. And we want to know about it. This kind of knowledge, contrary to instrumentalism, is and should be the goal of theorizing.

Underdetermination

2. The Position of the Empiricist:

- This is a kind of antirealism that claims that scientific theories are in fact true or false in what they say about both observables and unobservables, but we can have justified knowledge only about what they say about observables.
- The responsible position to take with regard to claims about what cannot be observed is a position of agnosticism: Withhold belief in those claims that are beyond the limits of knowledge.

Last time!

Underdetermination

- Grant science its aims of producing ***true*** accounts (descriptions) of both the observable causes and constituents (that's the *realist* position).

2. The Position of the Empiricist:

- The empiricist will do this but will also point out the problems of **underdetermination**:
  - ***All confirmation***, that is, all distinctions of the ***likely-to-be-true*** from the ***likely-to-be-false*** theories, ***will be seriously equivocal***, and testing against the evidence cannot distinguish any one of the unlimited number of empirically equivalent theories.
  - In addition, there is a problem with explanation: ***Explanation*** in general, judged by its contribution to our understanding of things, is a pragmatic and psychological event with ***no clearly truth-conducive features***.
  - If we specialize in strictly causal explanation, the evidence still underdetermines which causal explanation is the right causal explanation.

Last time!

Underdetermination

- Grant science its aims of producing ***true*** accounts (descriptions) of both the observable causes and constituents (that's the *realist* position).

2. The Position of the Empiricist:

- demands *agnosticism* towards unobservables.
- The idea is not to say that there are no such things as electrons or quarks. It is rather to say that we simply don't know and we in fact *can't* know if there are such things.
    - A claim about electrons or quarks is beyond the limits of justification and so beyond the limits of knowledge – the responsible attitude is to withhold judgement.
    - Though there is a good reason to use the theory ***as if it is true***, there is no good reason to *believe that **it is true***.

Underdetermination

3. Realism:

- assumes that the theories of science are in fact true or false of the world, or at least approximately so, and that we can sometime tell of a theory whether it is true or false.
- In other words, science can and does make justified knowledge claims about aspects of the world that cannot be observed.

UNDERDETERMINATION CONT.

Underdetermination

Reflect & Discuss

We have looked at how anti-realists justify their conservative position. On the other hand, the realist takes a bolder position.

- **How does the realist justify the link between empirical adequacy and truth of a theory?**

Underdetermination

3. Realism:

- Our best scientific theories (aim to) give true or approximately true descriptions of observable and unobservable aspects of the world.

- Scientific realism pivots on the ***distinction*** between ***empirical adequacy of a theory*** (something is directly testable) and **the *truth of a theory*** (a feature that is not directly testable).
- For a realist the two are linked; yet, a realist must accept the burden to demonstrate this link.

Underdetermination

3. Realism:

- The standard presentation of the link between empirical adequacy and truth of a theory is known as the **inference to the best explanation**.
    - Simply put, the realist claim is that there is good reason to believe in the truth of the theory that is the best causal explanation of the phenomena.
    - Identifying the ***best*** explanation limits the collection of acceptable theories to just one …

Underdetermination

Example: **Inference to the best explanation**

- Say you discover that your car won’t start in the morning and you would like to know ***why*** it won’t start (the explanation for the failure)
  - Possible explanations:
    - 1. The battery is dead.
    - 2. The fuel tank is empty.
    - 3. The starter has malfunctioned.
    - 4. A vandal has sabotaged the car.
    - 5. All or several of the above.

Underdetermination

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  - Possible explanations:
    - 1. The batt
    - 2. The fuel
    - 3. The start
    - 4. A vandal has sabotaged the car.
    - 5. All or several of the above.

No signs of tampering.
Eliminate 4.

Underdetermination

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    - 1. T
    - 2. The fuel tank is empty.
    - 3. The starter has malfunctioned.
    - ~~4. A vandal has sabotaged the car.~~
    - 5. All or several of the above.

Fuel gage says full.
Eliminate 2.

Underdetermination

Example: **Inference to the best explanation**

- Say you di Lights, heat, and radio work. Eliminate 1. start in the morning and you would like to know ***why*** it wo or the failure)
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Try to start and you hear a strange clicking sound. This now seems plausible.

Underdetermination

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  - Possible explanations:
    - 1. ~~The battery is dead.~~
    - 2. ~~The~~
    - 3. The … d.
    - 4. ~~A v … ar.~~
    - 5. All or several of the above.

A bunch have been ruled out. Eliminate 5.

Underdetermination

Example: **Inference to the best explanation**

- Say you discover that your car won't start in the morning and you would like to know ***why*** it won't start (the explanation for the failure)
  - Possible

    Best explanation!

    - 1. ~~T~~
    - 2. ~~T~~
    - 3. The starter has malfunctioned.
    - 4. ~~A vandal has sabotaged the car.~~
    - 5. ~~All or several of the above.~~

Underdetermination

3. Realism:

- The standard presentation of the link between empirical adequacy and truth of a theory is known as the **inference to the best explanation**.
  - Being an explanation is an external virtue since it involves comparison with observation, but being the best among empirically equivalent explanations will be a matter of being a ***simple, precise***, and ***fruitful theory*** that is ***compatible*** and ***cooperative with other acceptable theories.***

Underdetermination

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- The standard presentation of the link between empirical adequacy and truth of a theory is known as the **inference to the best explanation**.
  - Being an explanation is an external virtue since it involves comparison with observation, but being the best among empirically equivalent explanations will be a matter of being a ***simple, precise***, and ***fruitful theory*** that is ***compatible*** and ***cooperative with other acceptable theories.***
  - The *causal explanation that is most fruitful*, that is, *explains the most phenomena with a minimum of theoretical extravagance*, and that *fits into the larger theoretical network*, is ***more likely to be true*** than a theory that does none of this or does only some of it.

Underdetermination

Reflect & Discuss

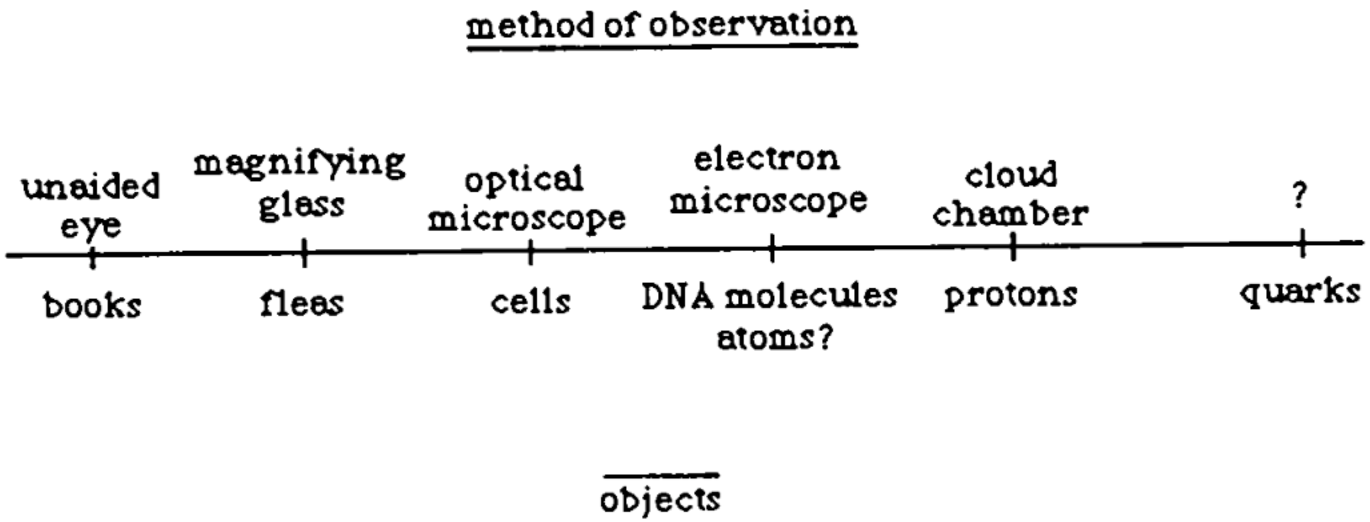
"The differences between realism and antirealism are largely over the significance of the distinction between observable and unobservable events and entities … We are justified (according to the antirealist view of science) to believe claims about observables, but for things like atomic theory and other claims about unobservables, there is at best justification to use the claim as if it is true."

- **What is the realist position on this matter?**

Underdetermination

3. The Realist View:

- sees no sharp limiting of justification by the factor of observability. Justification can cross the line of observability.
- will *undermine* the significance of observability by pointing out that entities do not fall neatly into one of the two categories (observable/unobservable); they occupy a more or less continuous spectrum of ***indirectness of observation***.

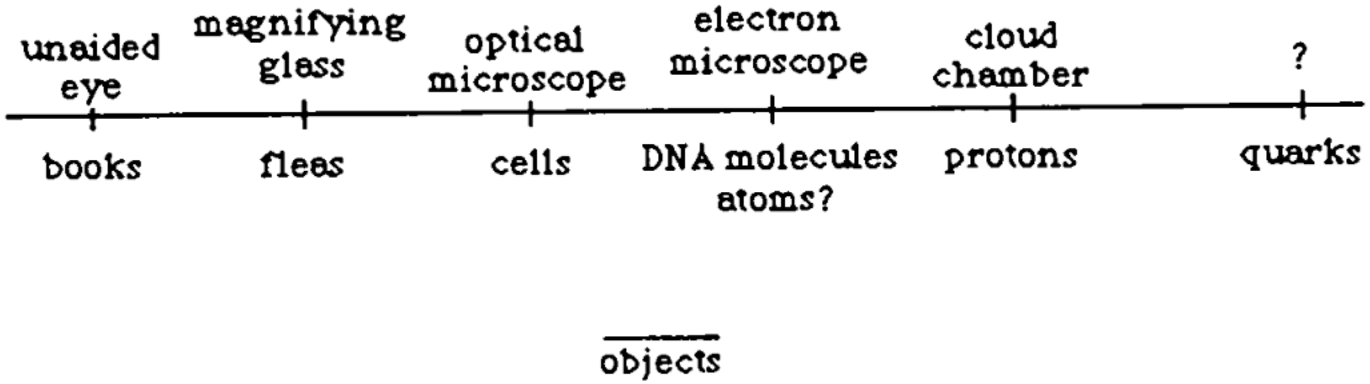


Underdetermination

3. The Realist View:

- sees n … fication can cross
- will *u* … es do not fall ne … occupy a more

The more apparatus necessary and the more inference used to sanction the observational claim, the more chance of misrepresenting the world.



Underdetermination

- The most profitable approach to understanding scientific knowledge and scientific method is to look for *degrees* of justification.
- Measurement of these degrees will be
    - in terms of internal virtues and external comparisons to evidence,
    - and it will even be sensitive to degrees of **observability** of the objects and events referred to by a theory.

OBSERVATION

OBSERVATION

- So far, we've been taking the idea of observation and observational claims as granted.

OBSERVATION

- Observation is the touchstone of ***external virtue*** – it is the contact of a theory to the outside world.
  - The underlying expectation – when we discussed explanation and confirmation – was that observation is *a source of objectivity in science*.

OBSERVATION

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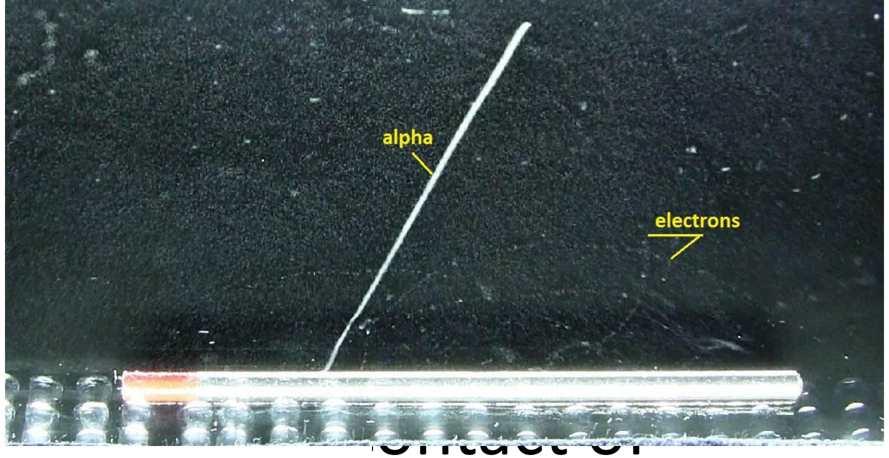
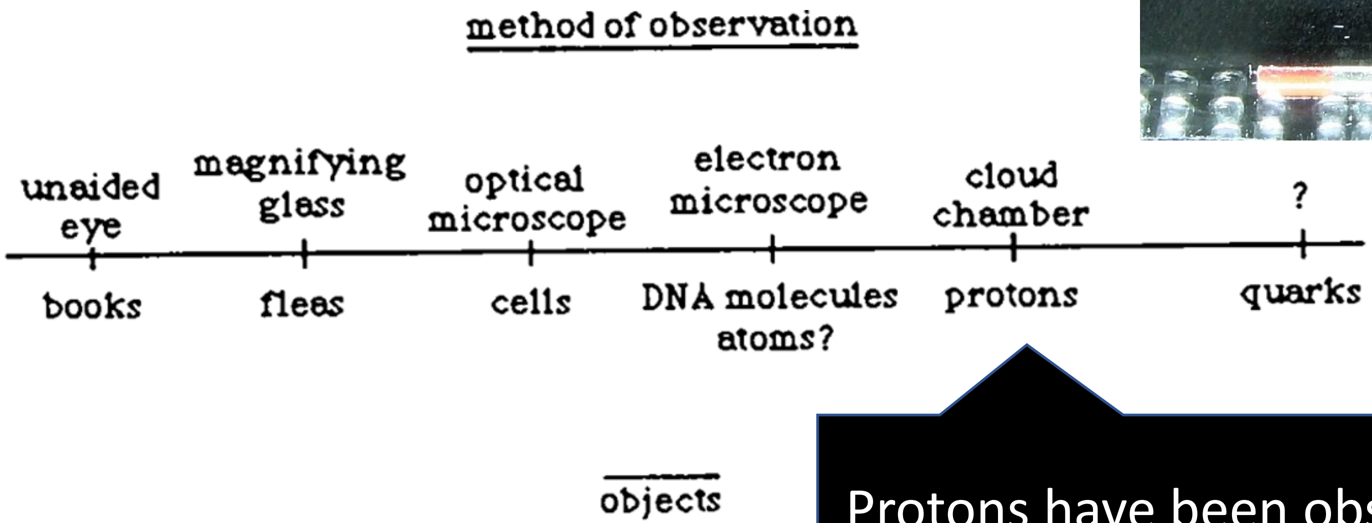
  - The underlying expectation – when we discussed explanation and confirmation – was that observation is *a source of objectivity in science*.

- ***However, this needs a closer investigation***
  - … we just saw earlier that there are degrees of observability parallel to degrees of indirectness in the observation!

OBSERVATION





- Observat
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  - The un
    confirn *ience*.

- ***However, this needs a closer investig***
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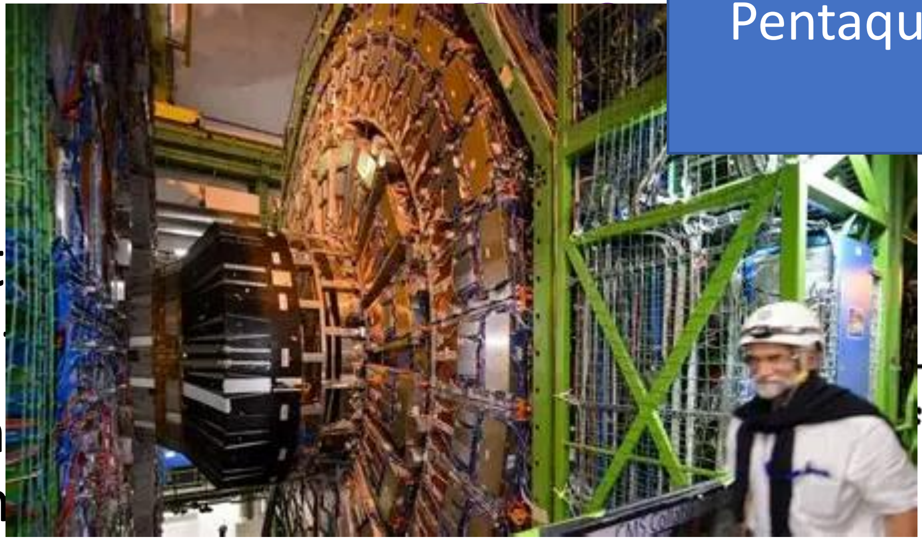
Protons have been observed (think of cloud chamber images)

Pentaquarks have been observed at CERN.



- Observat ... ontact of a theory ...
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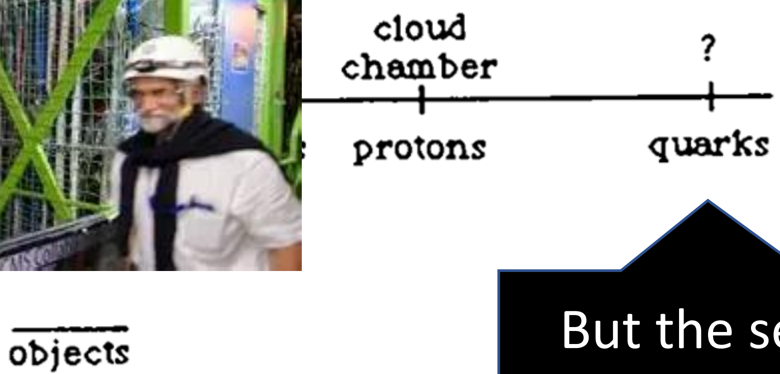

- ***However, this needs a closer investigation***
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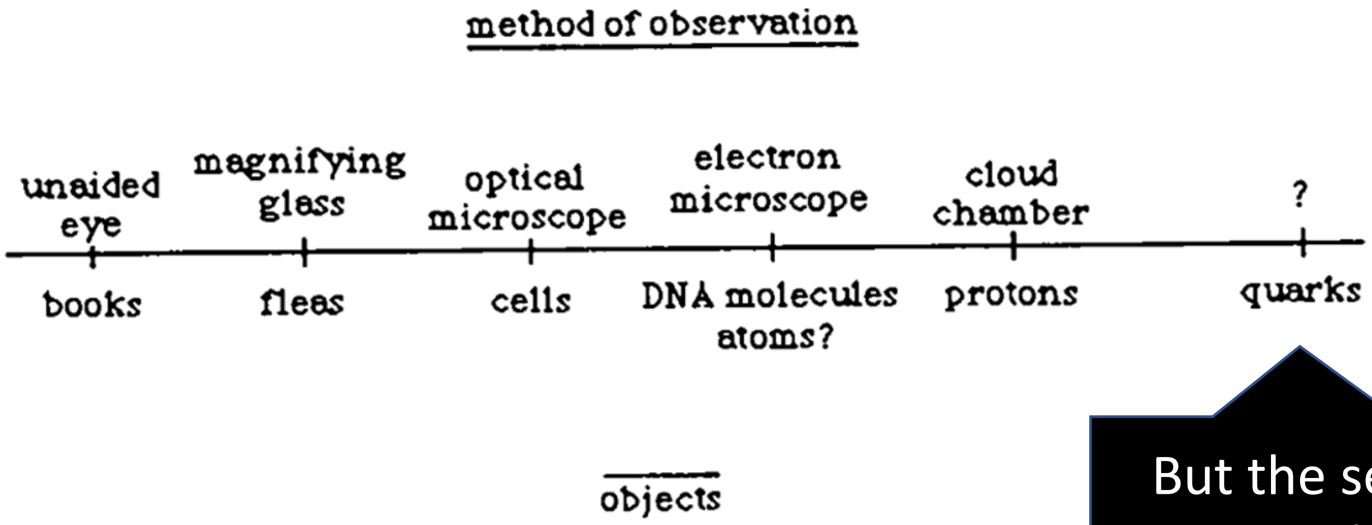
But the search for the top quark continues …
The search for ‘truth’ continues

OBSERVATION



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- ***However, this needs a closer investigation***
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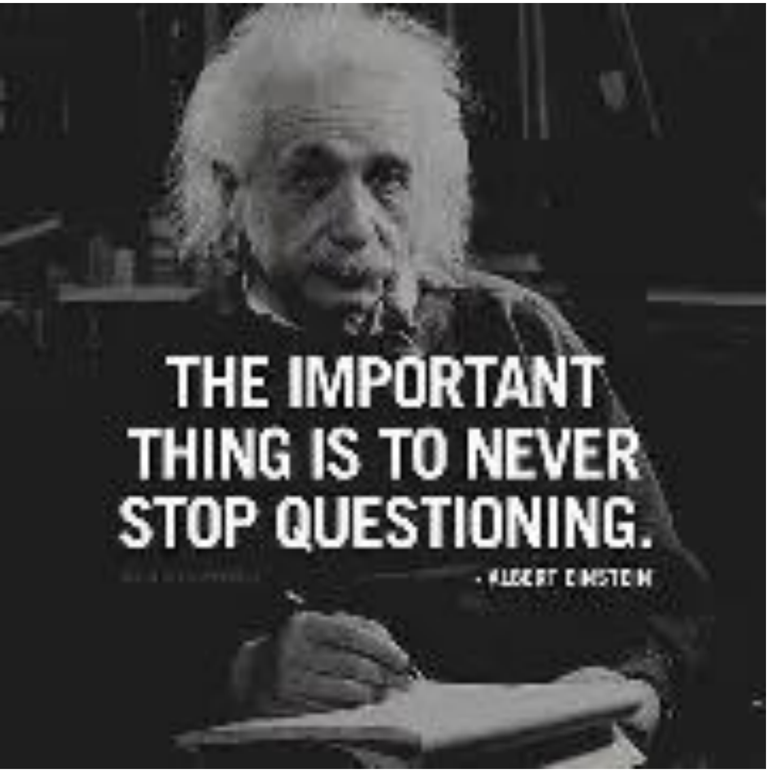
OBSERVATION

OBSERVATION

So, with the Hubble telescope we could see objects as they were at very nearly at the beginning of time.

OBSERVATION



- But are these examples of information (e.g., cloud chamber images, the Hubble telescope, etc.) straight from nature, free of any conceptual embellishment, the undeniable, impartial, common ground of objectivity?

Asking this question is important!
Why?

OBSERVATION

- Because we deal with *empirical sciences*: natural sciences (e.g., physics, biology) and social sciences (e.g., psychology and anthropology)
  - They all make claims on what is going on in the actual world (the world of elements, organisms, environments, etc.)
  - They contrast with claims in nonempirical sciences (e.g., mathematics, logic), which can be assessed without having to look at what's happening in the world.
    - For instance, I don't need to look outside in order to know that 2 + 2 = 4 or that a triangle has three sides.

OBSERVATION

- Reading ***the book of nature*** – a metaphor for doing natural science – requires observing the events in nature.

- To have any claim to an accurate report of what's going on in ***this*** world, there must be ***input*** from the world.

- Observation is, thus, a dietary staple of empirical science.

OBSERVATION

Reflect & Discuss

1. *Observations, being imposition from the world, are intersubjective and public in a way that beliefs and theories don’t have to be.”*

- **Explain this sentence in plain language.**
- **What is the assumed premise in this statement?**
- **Read the passage on Percepts and Constructs**

OBSERVATION

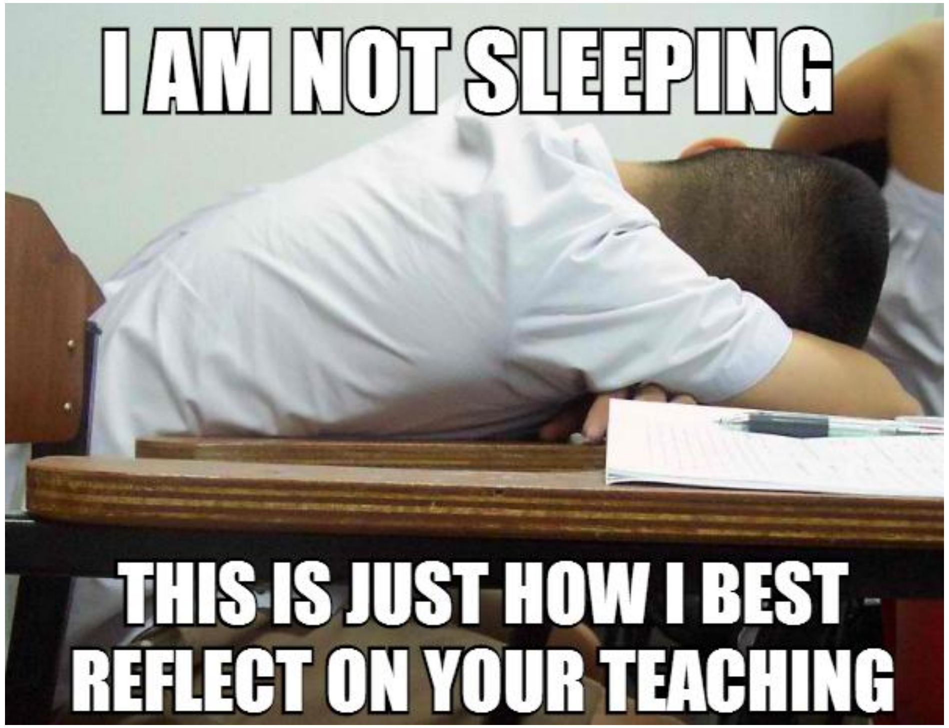
- The (empiricists') hope is that we all agree on observations since observations are accessible to all of us.
  - Even if we are skeptical about a theory or lean towards an alternative theory, we can all agree on the observations.

OBSERVATION

- The (empiricists') hope is that we all agree on observations since observations are accessible to all of us.
  - Even if we are skeptical about a theory or lean towards an alternative theory, we can all agree on the observations.
    - ***Uhhhh Ohhhh*** *so observations are free of inference and interpretations and, thus, there is no danger of error and/or bias?*

OBSERVATION

- Consider the following claims:
  - There are 15 people in the room.
  - One person is asleep.
  - 32 people have taken notes of the assigned readings.



These claims are easy to check, and we'll reach an agreement easily…

OBSERVATION

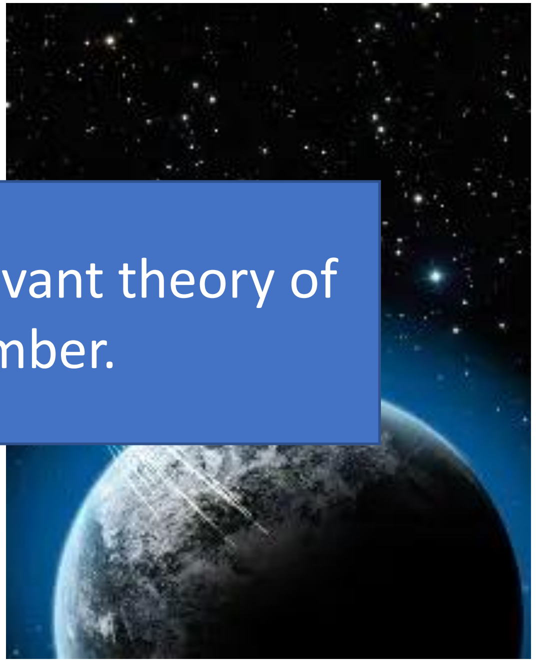
- But how about those claims:
  - There is at least one cosmic ray in the room at any one moment.
  - On average there are more cosmic rays going down than there are going horizontally.
  - There are fewer cosmic rays in the room on Sunday than on any other day.

Which of these questions can be checked and agreed on?

OBSERVATION

- But how about those claims:

To check these claims, we would need to consult the relevant theory of cosmic rays or use a device such as a cloud chamber.



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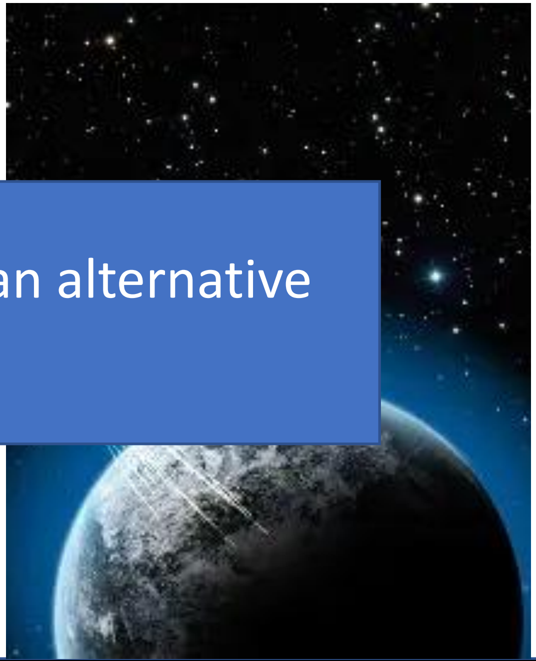

However, the agreement disappears as long as there is an alternative theory.

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Which of these questions can be checked and agreed on?

For example, one might suggest that the lines are formed by a magnetic effect and that what the cloud chamber shows are actually the magnetic field lines of the earth and nearby magnets …

OBSERVATION

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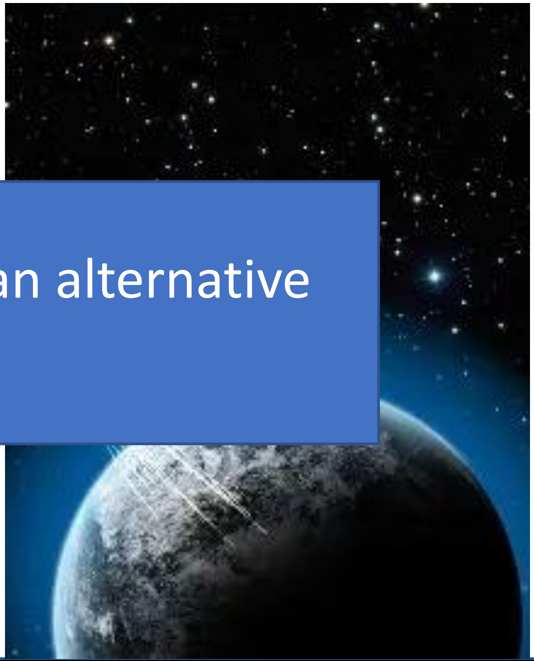

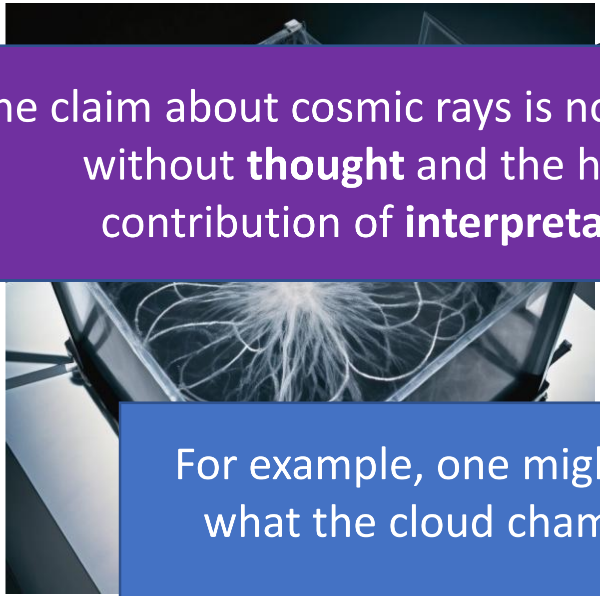

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The claim about cosmic rays is not decidable without **thought** and the human contribution of **interpretation**.

Which of these questions can be checked and agreed on?

For example, one might suggest that the lines are formed by a magnetic effect and that what the cloud chamber shows are actually the magnetic field lines of the earth and nearby magnets …

OBSERVATION

- Thus, the focus of our understanding of observation must be on observational report rather than on the physical event of sensation.

OBSERVATION

Reflect & Discuss

2. "To serve as justification, an observation must have some assertive content …"

- **Explain**

3. "the observation report must itself be justified if it is to have sufficient assertive authority to prove other claims."

- **What factors are necessary in order to justify an observational claim?**

OBSERVATION

- **Accountable Observation:**
  - To be useful as an evidence, an observation must make contact with a theory in some way if it is to serve as proof or as refutation.
  - Science can make use of observation only if it is an accountable observation:
    1. The observation must be *informative* in the sense that it must be an account *of* something if it is to have the assertive content required to serve as justification of other claims.
       - ➢ It requires a claim as to ***what*** is observed (*What do you see*?) and an assertion of the ***information*** that is being conveyed to the world (*Do you see that this is a that?*)
       - ➢ If an observation report is to be relevant to a theory it must include some of the language of the theory.
       - ➢ It must be assertively be evidence of an x if it is to be evidence in the case to justify a theory about x.

OBSERVATION

- **Accountable Observation:**
  - To be useful as an evidence, an observation must make contact with a theory in some way if it is to serve as proof or as refutation.
  - Science can make use of observation only if it is an accountable observation:
    1. The observation must be *informative* in the sense that it must be an account *of* something if it is to have the assertive content required to serve as justification of other claims
    2. And the observation itself must be justified in the sense of being certifiably not haphazard or uncontrolled.
       - ➢ Observations must be *carefully* done and *repeatable*; carried out in a *controlled way and under proper conditions.*
       - ➢ This requires an understanding of the relevant conditions under which an observation is done.
         - ➢ The scientific community must understand what it takes to do the observation properly if the observation is to be acceptable as scientific evidence.

OBSERVATION

- **Accountable Observation:**
  - To be useful as an evidence, an observation must make contact with [illegible] ome way if it is to serve as proof or as refutation.
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[illegible] must be *carefully* done and *repeatable*; carried out in a *controlled way and under* [illegible] *conditions.*

This requires an understanding of the relevant conditions under which an observation is done.

➢ The scientific community must understand what it takes to do the observation properly if the observation is to be acceptable as scientific evidence.

In short: To contribute to science, an observation must be informative, assertive, propositional.

OBSERVATION

Reflect & Discuss

4. "All observation in science is influenced by theory. This is not simply the way things happen to be done; it is the way things have to be."

- **In what way does theory influence observation?**

- **What are accounting theories and what is their significance?**

OBSERVATION

- All observation in science is influenced by theory:
  - Informative observation is impossible without a *theoretical background*, and
  - *Alternative theoretical backgrounds* lead to *alternative informational reports*.

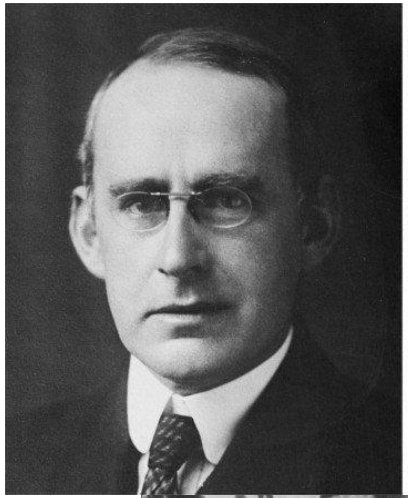
Observation without theory is blind!

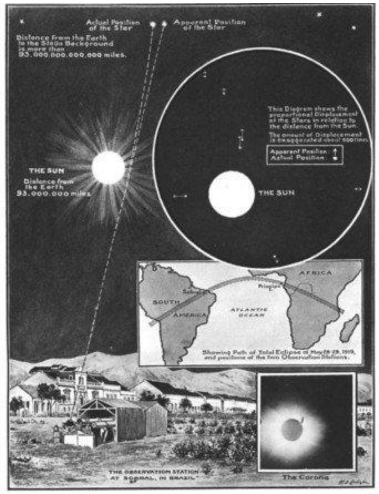
OBSERVATION

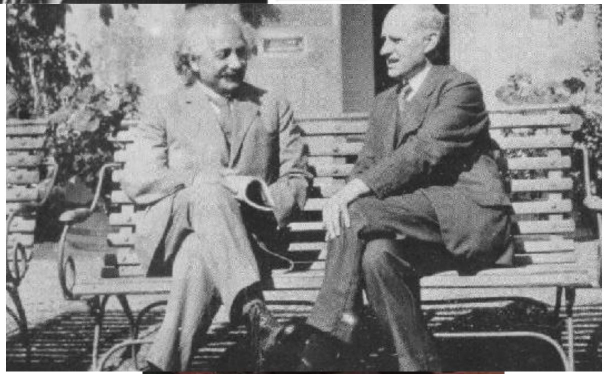
- Theory influences observation in several ways:
  - ❖ Theoretical understanding of the world guides our ***selection*** of ***what*** observations to do.

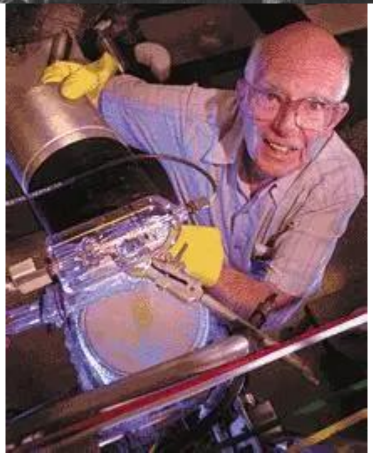
    Example:
    - Einstein’s theory indicated that an observation of stars during an eclipse would be informative.
    - Solar theory directs observational efforts toward the search for neutrinos.
  - ➢ This is, of course, the way things have to be and ought to be if the idea is to test a theory by its consequences.

❖Different theories precipitate a different collection of evidence.









OBSERVATION

- Theory influences observation in several ways:
  - Theoretical understanding of the world guides our ***selection*** of ***what*** observations to do.
  - Theories guide our decisions about when to ***count*** observations as ***credible***.
    - Theories supply standards to *evaluate the reliability* of observational reports (e.g., the assessment of the viewing conditions, such as adequate lighting; attesting that there are no distorting factors, etc.)

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      - o The assessment of the viewing conditions, such as adequate lighting; attesting that there are no distorting factors, etc., is largely based on an understanding of the *causal mechanisms* of observation.
      - o The role of theory is especially apparent in cases of observations that are assisted by machines.
        - o *Example*: Telescopes, if properly constructed, do not manufacture false images – the accuracy of the image is sanctioned by a theory of optics.

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      - o The role of theory is especially apparent in cases of observations that are assisted by machines.
        - o A great amount of theory goes into the building of the machines (e.g., telescope, microscope, cloud chamber), and into sanctioning their reliability.
        - o The data is acceptable, that means, the observations are accountable, only if the experts, who invent and build the machines, have a solid theoretical account of *how they work.*

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Accounting theory:

A theoretical claim used to attest to the reliability of an observation or to describe the informational content of an observational claim is being used as an accounting theory.

It is a designation of how a theory is used rather than of any intrinsic features of content or scope of a theory.

As an example, theoretical claims about the behavior of electrons are used as accounting theories in the case of observing a DNA molecule (or anything else) with an electron microscope.

OBSERVATION

Reflect & Discuss

Read the passage on Theory-Laden Facts

5. Describe the analogy between reading a text and doing science, while connecting these activities to the ideas in the passage on Theory-Laden Facts.

OBSERVATION

- Observations in science are theory-laden
  - They always depend on the theories we use to *interpret* them.
    - *Example*: To say that the "Sun rises in the east every day", involves an understanding of the concept of "Sun" and "east" as well as geographical and astronomical knowledge.
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  - This dashes the empiricist's hope of a common ground of evidence agreeable to everyone regardless of his or her theoretical predisposition.
- ***Observations are beliefs we get as a result of direct impact by the environment, though they gain their importance as evidence to function in the process of science only with reference to theory.***

OBSERVATION

- Reading ***the book of nature*** – a metaphor for doing natural science – requires observing the events in nature.
- To have any claim to an accurate report of what's going on in ***this*** world, there must be ***input*** from the world.
- Observation is, thus, a dietary staple of empirical science.

OBSERVATION

- When we read a book, we want to know the plot and the meaning of the text
  - We want to learn what is beyond the manifest *marks* on the pages – we want to understand what the *marks* indicate and what they mean.

OBSERVATION

- When we read a book, we want to know the plot and the meaning of the text
  - We want to learn what is beyond the manifest *marks* on the pages – we want to understand what the *marks* indicate and what they mean.
- Science is doing the same!
  - Science aims to understand what the *observations* indicate, and this is done only through a careful attention to the observations themselves.
    - This includes forming concepts of things so that tokens of sensation can be grouped into interacting kinds of things and the relevant kinds can be individuated and understood.
  - Thus, observations are meaningful contributions to our understanding of nature only with some claims of connection between the image in sensation and the specimen in the world.

OBSERVATION

- Reading ***the book of nature*** – a metaphor for doing natural science – requires observing the events in nature.

- To translate the ***book of nature*** one needs *theory* of *what the observations mean*, a *lexicon of what the marks are evidence of*.